Exercise 1

1-1. Consider the circuit of Fig.1.1.

a) Using the simple model with V_{Don} = 0.7V, solve for I_D;

b) Find I_D and V_D using the ideal diode equation. Use I_S =10^{-14} A and T=300 K.

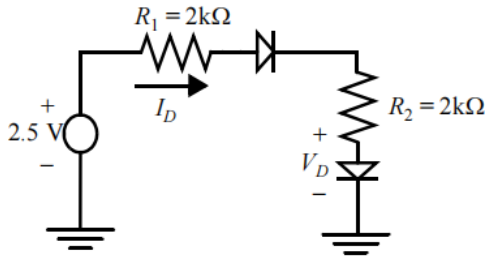


Fig.1.1

Answer:

a. $I_D = \frac{V_{in}-2*V_{Don}}{R_1+R_2} = \frac{2.5-2*0.7}{4K\Omega} = 275uA$

b. $I_S = 10^{-14}A, T = 300K, V_{Don} = 0.7V. I_D = I_S \times (e^{\frac{V_{Dq}}{KT}} - 1)$ where $\frac{KT}{q} = 26mV@300K$

$I_D = \frac{V_{in}-2\cdot V_{Don}}{R_1+R_2} = I_S \times (e^{\frac{V_{Dq}}{KT}} - 1)$ iterating on this expression we can obtain $\frac{V_D}{I_D} = \frac{0.628V}{311uA}$

1-2. For the circuit in Fig.1.2, V_s=3.3 V. Assume A_D = 12 μm^2, φ_0 = 0.65 V, and m= 0.5, N_A = 2.5×10^{16} and N_D = 5×10^{15}.

a) Is the diode forward- or reverse-biased?

b) Find I_D and V_D;

c) Find the depletion region width, W_j, of the diode;

d) Use the parallel-plate model to find the junction capacitance, C_j;

e) Set V_s = 1.5 V. Again using the parallel-plate model, explain qualitatively why C_j increases.

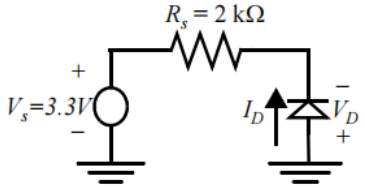


Fig.1.2

Answer:

a. Reverse biased

b. $I_D = I_{REV} = -I_S \approx 0, V_D = -V_S = -3.3V$

c. $W_j = \sqrt{\frac{2\cdot\varepsilon_{si}}{q} \times \frac{N_A+N_D}{N_A N_D} \times (\varphi_0 - V_D)}, q = 1.6\times10^{-19}C, V_D = -V_S = -3.3V,$

$\varepsilon_{si} = 11.7\varepsilon_0 = 1.035\times10^{-10}\,{}^{F}/_{m}, N_A = 2.5\times\frac{10^{16}}{\mathrm{cm}^3}, N_D = 5\times\frac{10^{15}}{\mathrm{cm}^3}.$

$W_j = 1.107\times10^{-4}cm.$

d. $C_j = \frac{\varepsilon_{si}\cdot A_D}{W_j}, A_D = 12\times10^{-8}cm^2, C_j = 1.12\times10^{-15}F.$

e. V_{SNEW} = 1.5V < V_{SOLD} = 3.3V. The new voltage reduces the reverse bias of the PN junction , hence the width of the depletion region ,W_j, decreases. As you bring the plates of capacitor together ,the capacitance increases.

1-3. Fig.1.3 shows NMOS and PMOS devices with drains, source, and gate ports annotated. Determine the operation region (saturation, linear, or cut-off) and the drain current I_D for each of the biasing configurations given in table. Assume the model parameters from Table.1.1, V_{BS}=0 and W/L = 1, L=1um, fill the table

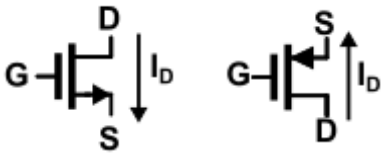


Fig.1.3

Answer:

	V_{GS}(V)	V_{DS}(V)	Operation region	I_D
NMOS	2.5	2.5	saturation	392.04uA
	3.3	2.2	linear	726uA
	0.6	0.1	cut-off	0
PMOS	-0.5	-1.25	cut-off	0
	-2.5	-1.8	saturation	176.58uA
	-2.5	-0.7	linear	101.5uA

NMOS :1. Saturation $I_D = k_n'\frac{W}{L}(V_{GS}-V_{TH})^2(1+\boldsymbol{\lambda}V_{DS}) = 392.04uA;$

2. linear $I_D = 2\times k_n'\frac{W}{L}\left((V_{GS}-V_{TH})V_{DS} - \frac{{V_{DS}}^2}{2}\right) = 726uA;$

3. cut-off $I_D = 0,$

PMOS : 1. cut-off $I_D = 0,$

2. Saturation $I_D = k_p'\frac{W}{L}(V_{GS}-V_{TH})^2(1+\boldsymbol{\lambda}V_{DS}) = 176.58uA$

3. linear $I_D = 2\times k_p'\frac{W}{L}\left((V_{GS}-V_{TH})V_{DS} - \frac{{V_{DS}}^2}{2}\right) = 101.5uA$

1-4. An NMOS device is plugged into the test configuration shown below in Fig .1.4 The input V_{in} is 2V. The current source draws a constant current of 50 μA. R is a variable resistor between 10kΩ and 30 kΩ. Transistor M1 has following transistor parameters: k'= 110μ V/A^2, V_T = 0.7V, and V_{DSAT} = 0.6V, and has a W/L = 2.5μm/0.25um. For simplicity, the body effect and channel length modulation can be neglected, i.e λ=0, γ=0.

a) When R =10kΩ find the operation region, V_D and V_S.

b) For the case of R = 10kΩ, would V_S increase or decrease if $\lambda \neq 0$. Explain qualitatively.

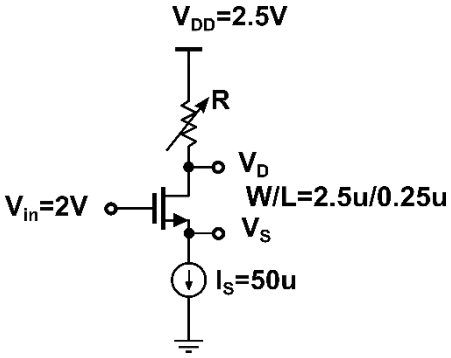


Fig.1.4

Answer:

a. When R =10K, V_D=V_{DD}-IR=2.5-50 $\times 10^{-6} \times 10^4$=2.5-0.5=2$V$.Assume the device is in saturation $I_D = k'_n \frac{W}{L}(V_{GS} - V_{TH})^2 = 50uA$ find $V_{GS} - V_{TH} = 0.213V$,so $V_{GS} = 0.213 + 0.7V = 0.913V$,

$V_S = 1.087V$.V_{GS}=0.913V,V_{DS}=1.087V device in the saturation.

b. Increase. V_D is fixed due to constant current.$1 + \lambda V_{DS}$ form would try to increase the current more then 50uA, thus V_{GS} needs to reduce by increase V_S.

Thinking Questions(optional)

1-5. Show that two MOS transistors connected in parallel with channel widths of W_1 and W_2 and identical channel lengths of L can be modeled as one equivalent MOS transistor whose width is W_1+W_2 and whose length is L, as shown in Fig.1.5 Assume the transistors are identical except for their channel widths.

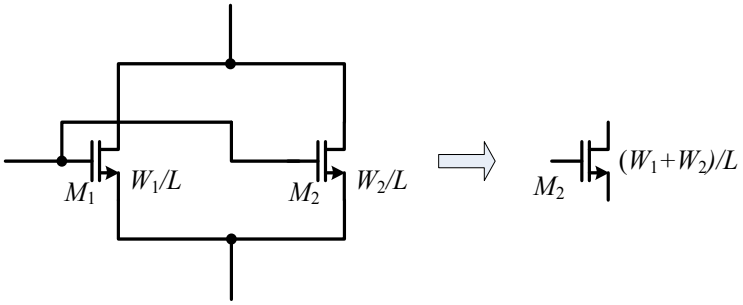


Fig.1.5

Answer:

For $V_{DS} \le V_{GS} - V_{TH}$

$$I_{D1} = \mu C_{OX} \frac{W_1}{L}[(V_{GS} - V_{TH})V_{DS} - \frac{1}{2}V_{DS}^2]$$

$$I_{D2} = \mu C_{OX} \frac{W_2}{L}[(V_{GS} - V_{TH})V_{DS} - \frac{1}{2}V_{DS}^2]$$

……

$$I_{Dn} = \mu C_{OX} \frac{W_n}{L}[(V_{GS} - V_{TH})V_{DS} - \frac{1}{2}V_{DS}^2]$$

$$\therefore I_D = I_{D1} + I_{D2} + \ldots + I_{Dn} = \mu C_{OX} \frac{W_1 + W_2 + \ldots + W_n}{L}[(V_{GS} - V_{TH})V_{DS} - \frac{1}{2}V_{DS}^2]$$

For $V_{DS} \ge V_{GS} - V_{TH}$

$$I_D = I_{D1} + I_{D2} + \ldots + I_{Dn} = \frac{1}{2}\mu C_{OX} \frac{W_1 + W_2 + \ldots + W_n}{L}(V_{GS} - V_{TH})^2$$

Thus the equivalent length = L and the equivalent width = $W_1 + W_2 + \ldots + W_n$.

1-6. Show that two MOS transistors connected in series with channel lengths of L_1 and L_2 and identical channel widths of W can be modeled as one equivalent MOS transistor whose width is W and whose length is L_1+L_2, as shown in Fig. 1.6. Assume the transistors are identical except for their channel lengths. Ignore the body effect and channel-length modulation.

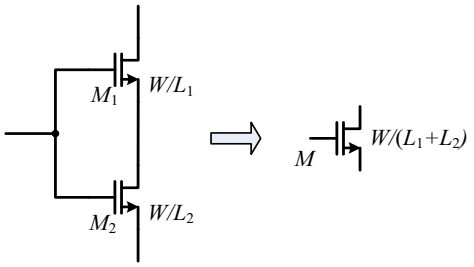
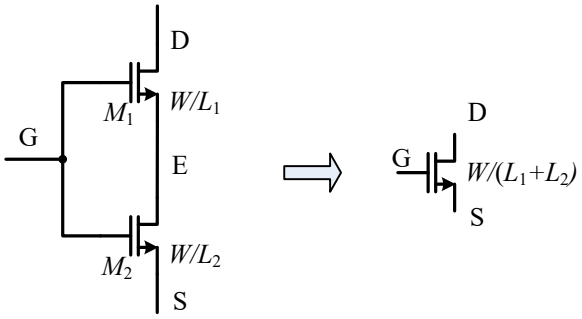


Fig.1.6

Answer:



(1) When $V_{GS}<V_{TH}$ and $V_{GE}<V_{GS}<V_{TH}$, the MOSFETs are in cut off.

(2) While M1 operates in triode ($V_{DE} < V_{GE} - V_{THN}$), that is equivalent to $V_{DE} + V_{ES} < V_{GE} + V_{ES} - V_{THN}$, i.e. $V_{DS} < V_{GS} - V_{THN}$.

Thus M2 operates in triode, too.

Thus

$$I_{D1} = \mu_n C_{OX} \frac{W}{L_1}[(V_{GE} - V_{TH})V_{DE} - \frac{1}{2}V_{DE}^2] \quad (1)$$

$$I_{D2} = \mu_n C_{OX} \frac{W}{L_2}[(V_{GS} - V_{TH})V_{ES} - \frac{1}{2}V_{ES}^2] \quad (2)$$

Since

$$V_{DS} = V_{DE} + V_{ES} \quad (3)$$

$$V_{GE} = V_{GS} - V_{ES} \quad (4)$$

$$I_{D1} = I_{D2} = I_D \quad (5)$$

It can be derived from equations (1), (2), (3), (4) and (5) that

$$(V_{GS} - V_{TH})V_{ES} - \frac{1}{2}V_{ES}^2 = \frac{L_2}{L_1}[(V_{GS} - V_{TH} - V_{ES})(V_{DS} - V_{ES}) - \frac{1}{2}(V_{DS} - V_{ES})^2]$$

$$= \frac{L_1}{L_1 + L_2}[(V_{GS} - V_{TH})V_{DS} - \frac{1}{2}V_{DS}^2]$$

So we can get $I_D = \mu_n C_{OX} \frac{W}{L_1 + L_2}[(V_{GS} - V_{TH})V_{DS} - \frac{1}{2}V_{DS}^2]$

(3) While M1 operates in saturation ($V_{DE} > V_{GE} - V_{THN}$). It means $V_{DE} + V_{ES} > V_{GE} + V_{ES} - V_{THN}$, i.e. $\boldsymbol{V_{DS} > V_{GS} - V_{THN}}$.

$V_E = V_G - V_{GE} < V_G - V_{THN}$, it means $\boldsymbol{V_{ES} < V_{GS} - V_{THN}}$. M2 operates in triode.

So

$$I_{D1} = \frac{1}{2}\mu_n C_{OX} \frac{W}{L_1}(V_{GE} - V_{TH})^2 \quad (1)$$

$$I_{D2} = \frac{1}{2}\mu_n C_{OX} \frac{W}{L_2}[2(V_{GS} - V_{TH})V_{ES} - V_{ES}^2] \quad (2)$$

$$V_{DS} = V_{DE} + V_{ES} \quad (3)$$

$$V_{GE} = V_{GS} - V_{ES} \quad (4)$$

$$I_{D1} = I_{D2} = I_D \quad (5)$$

It can be derived from equations (1), (2), (3), (4) and (5) that

$$I_D = \frac{1}{2}\mu_n C_{OX}\frac{W}{L_1 + L_2}(V_{GS} - V_{TH})^2$$

That just like a MOSFET operating in saturation, which has a length of L_1+L_2 and a width of W. It can be deducted similarly that n MOSFETs in series acts as a MOSFET with an aspect ratio of W/(L_1+L_2+…L_n).

Table.1.1

Parameter Symbol	Parameter Description	Typical Parameter Value		Units
		n-Channel	p-Channel	
V_{T0}	Threshold voltage (V_{BS}=0)	0.7	－0.7	V
K'	Transconductance parameter (in saturation)	110.0	50.0	μA/V^2
γ	Bulk threshold parameter	0.4	0.57	$V^{1/2}$
λ	Channel length modulation parameter	0.04 (L=1μm) 0.01 (L=2μm)	0.05 (L=1μm) 0.01 (L=2μm)	V^{-1}
$2\|\Phi_F\|$	Surface potential at strong inversion	0.7	0.8	V

$$*K' = \frac{1}{2}\mu C_{ox}$$